Artificial Intelligence in Physics Education: Opportunities for Students with Disabilities

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Introduction

Physics is one of the most conceptually demanding disciplines in STEM, requiring simultaneous integration of mathematical reasoning, conceptual abstraction, and technical reading [1]. These overlapping demands create significant barriers for students with disabilities.

KEY BARRIERS IDENTIFIED:

- Rigid assume standardized learning modes, disadvantaging neurodiverse students [1]
- High-stakes, time-pressured exams disproportionately impact students with ADHD, dyslexia, or dyscalculia and ASD [9]
- Dense technical textbooks present sustained reading challenges [7]
- Social stigma discourages students from disclosing disabilities or seeking accommodations [2]

Challenges faced by students with disabilities

Disability	Read	Math	Attn	Abstract
Dyslexia	✓		✓	✓
Dyscalculia		✓		✓
ADHD	✓		✓	
ASD	✓		✓	✓

Table 1: **Note:** ✓ indicates an area of difficulty.

Structural Challenges

How is AI used in Physics Education

FOR INSTRUCTORS:

- Generates multi-media content [12]
- Create interactive simulations of physical principles [12]
- Automated feedback on assignments [12]
- Analyze class performance data [20]

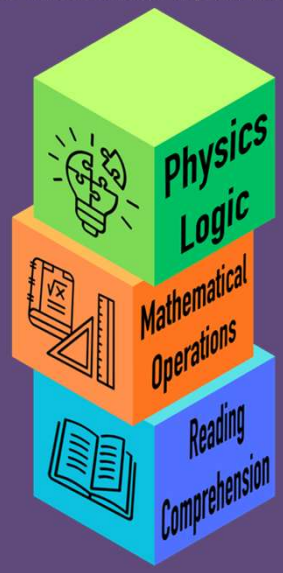
FOR STUDENTS:

- AI tutors decompose multi-step problems into manageable steps [21]
- On-demand, private, nonjudgmental support — no need to disclose disability [21]
- Adaptive pacing: slower for difficult concepts, faster for mastered material [21]
- LLMs summarize dense readings and clarify unfamiliar terminology [4[

How can AI be utilized to support students with disabilities in PhysEd?

AI FOR ACCESSIBILITY:

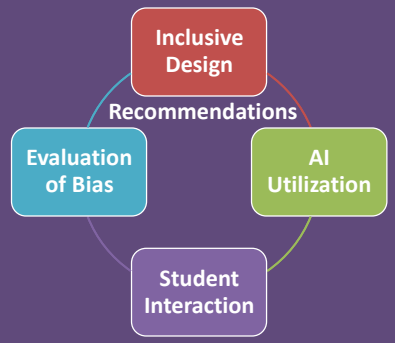
- Dynamic visualizations of quantum wavefunctions, fields, and waves reduce reliance on text [4]
- Iterative explanation loops let students find their preferred representation [4]
- Cognitive scaffolding reduces working memory load for multi-step derivations [8]
- AI writing assistants help structure lab reports and scientific explanations [4]



THE "BLACK BOX" PARALLEL:
Many disabled students experience their own cognition as opaque — AI systems that provide iterative, flexible feedback can interact productively with diverse cognitive styles without requiring students to verbalize their learning needs. [4,21]

What should we be concerned about?

- **Overreliance:** AI answers may reduce incentive to engage deeply with difficult material [7]
- **Hallucinations:** AI-generated physics solutions can contain subtle but critical errors [12]
- **Algorithmic bias:** Tools trained on narrow datasets may underserve neurodivergent users [4]
- **AI detection systems:** False positives disproportionately penalize students using assistive technologies [4]



Full Paper and Key References

[1] Chasen et al., Journal of Engineering Education, 2025.
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[7] Paglialunga & Melogno, Brain Sciences, 2025.
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[9] Almufareh et al., Universal Access in the Information Society, 2025.
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[20] Liang et al., Physics Education under the Application of AI, 2026.
[21] Khowaja & Salim, Science Journal of Education, 2025.